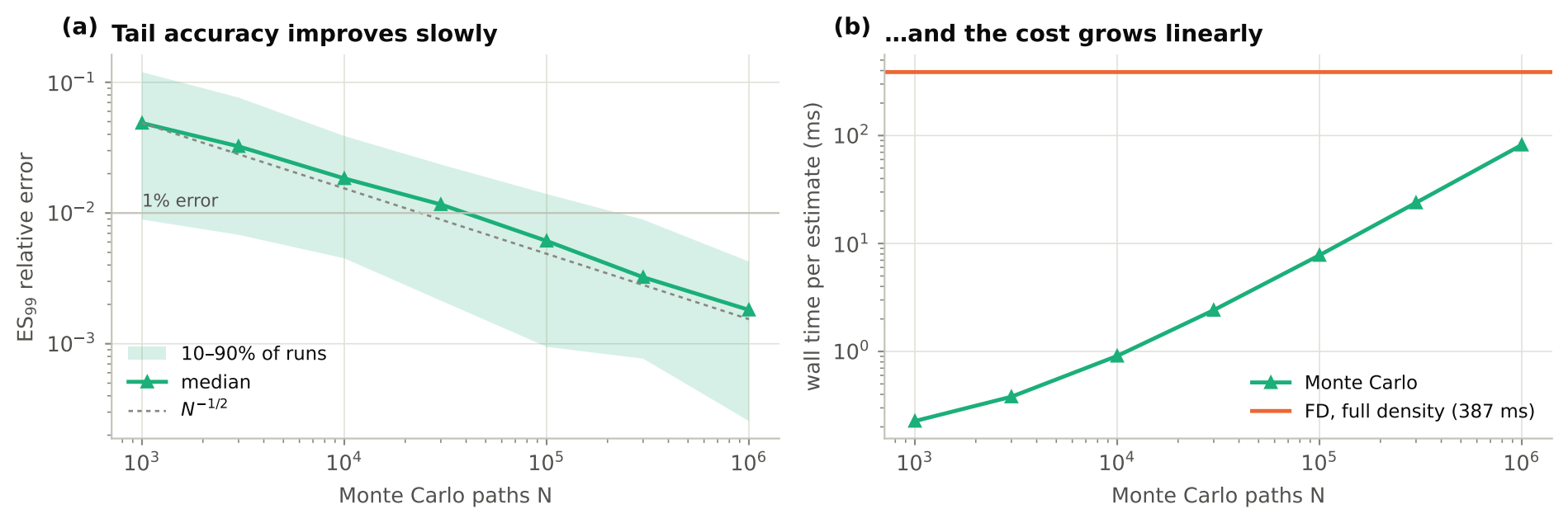
(a) Tail accuracy improves slowly
10^{-1}
10^{-2}
10^{-3}
ES_{99} relative error
1% error
10–90% of runs
median
$N^{-1/2}$
10^3
10^4
10^5
10^6
Monte Carlo paths N
(b) ...and the cost grows linearly
10^2
10^1
10^0
wall time per estimate (ms)
Monte Carlo
FD, full density (387 ms)
10^3
10^4
10^5
10^6
Monte Carlo paths N